Q1

1.1

- Using PyMOL, we defined the ACE2–RBD interface with a distance rule: selecting all ACE2 atoms within **4.0 Å** of any RBD atom and all RBD atoms within **4.0 Å** of ACE2, followed by taking the union of the corresponding residues as the potential interface *(Figure 1 A, B)*.
- Next, we compared these residues with hydrogen-bond and salt-bridge pairs reported in the **original essay**'s Extended Data Table 2 (Lan *et al.*, 2020).
- All literature residues were contained within the distance-based set, which supports the validity of this contact region.
- Complete ACE2 and RBD interface residues are summarized in *Table 1*.

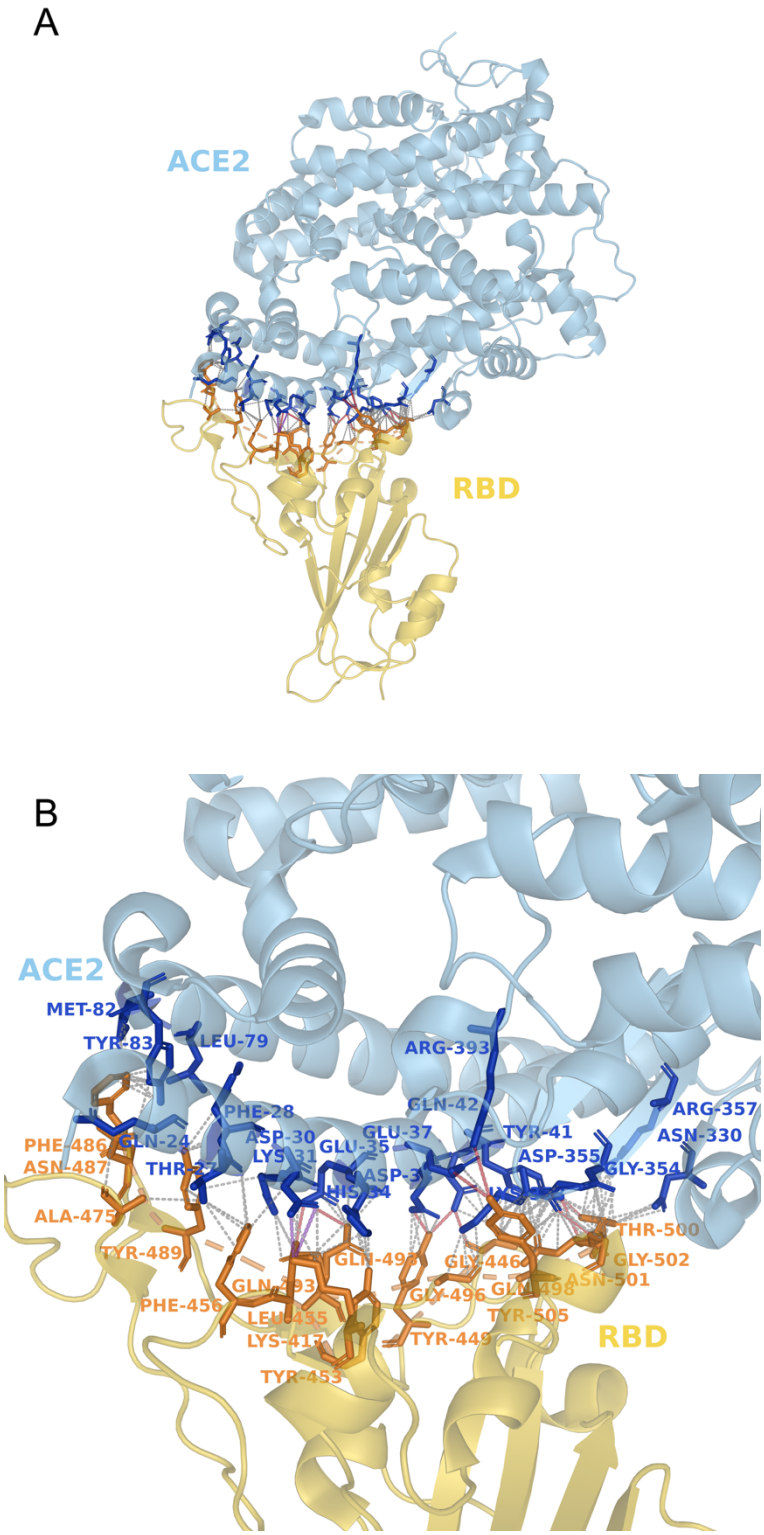


Figure 1. *Experimental ACE2–Spike RBD complex (PDB 6M0J) and interface. ACE2 (chain A) is shown as a light-blue cartoon and the RBD (chain E) as a yellow cartoon. Interface residues on ACE2 are drawn as dark-blue sticks, and the matching residues on the RBD are drawn as orange sticks.* ***(A)*** *Overall view of the complex.* ***(B)*** *Close-up view of the interface. Grey dashed lines connect atom pairs within 4.0 Å. Hydrogen bonds reported in the literature are highlighted in red, and salt bridges in purple.*

Interface Residues	
ACE2 (20 residues)	***RBD (17 residues)***
GLN-24	*LYS-417*
THR-27	*GLY-446*
PHE-28	*TYR-449*
ASP-30	*TYR-453*
LYS-31	*LEU-455*
HIS-34	*PHE-456*
GLU-35	*ALA-475*
GLU-37	*PHE-486*
ASP-38	*ASN-487*
TYR-41	*TYR-489*
GLN-42	*GLN-493*
LEU-79	*GLY-496*
MET-82	*GLN-498*
TYR-83	*THR-500*
ASN-330	*ASN-501*
LYS-353	*GLY-502*
GLY-354	*TYR-505*
ASP-355	
ARG-357	
ARG-393	

Table 1. ACE2 and RBD residues in the potential contact region identified from the 6M0J experimental structure using a 4.0 Å distance cutoff.

1.2

- ***Docking Results***
- Chain A of 6M0J (ACE2) was used as the receptor and chain E (RBD) as the ligand for protein–protein docking with ClusPro
- Under the **balanced scoring scheme**. The five top-ranked clusters, along with their corresponding weighted scores, are summarized in *Table 2*.
- ***Align & Comparison***
- The highest-ranked model was superimposed onto the experimental ACE2–RBD complex **using ACE2 as the reference** *(Figure 2 A)*.
- The two complexes clearly position the RBD in different regions of the ACE2 surface, indicating a significant discrepancy in interface location.
- To quantify this mismatch, the Cα RMSD was computed between the experimental and docked interfaces, using the 37 ACE2 residues defined in *Table 1*. The interface **Cα RMSD was 37.917 Å**, showing that the predicted orientation **shows little resemblance** to the crystallographic interface.
- This poor agreement highlights the limited accuracy of the present docking run and reinforces that reliable protein–protein docking remains a difficult and still challenging problem
- ***Model Structure Details***
- The docked interface was then analyzed to identify factors that could stabilize the model *(Figure 2 B)*.
- First, a **4.0 Å** distance cutoff was applied to scan the contact region and select potential **Van Der Waals interactions**.
- Second, **hydrogen bonds** were searched using a **3.2 Å** cutoff between donor and acceptor atoms. Additionally, the hydrogen bonds are highly consistent with those detected by **PDBePISA** in question 1.3.
- Third, **electrostatic complementarity (salt bridge)** was probed by examining interactions between oppositely charged side chains **(Arg/Lys/His vs. Asp/Glu)** within **10 Å**. No clear salt bridges were detected in this simple residue-pair analysis, suggesting that a more rigorous approach based on the overall surface charge distribution would be required.
- Finally, we identified residues contributing to BSA by comparing the per-residue monomer ASA to the complex ASA. Residues with **BSA values greater than 1.0 Å²** were considered to support the interface.

Cluster	*Members*	*Representative*	*Weighted Score*
0	*55*	*Center*	*-621.1*
		Lowest Energy	*-748.2*
1	*42*	*Center*	*-616.2*
		Lowest Energy	*-708.9*
2	*39*	*Center*	*-728.7*
		Lowest Energy	*-728.7*
3	*32*	*Center*	*-675.9*
		Lowest Energy	*-675.9*
4	*27*	*Center*	*-740.8*
		Lowest Energy	*-740.8*

Table 2. *Five best docking clusters and their weighted scores for ACE2 (receptor)–RBD (ligand) docking generated by ClusPro using the balanced scoring strategy.*

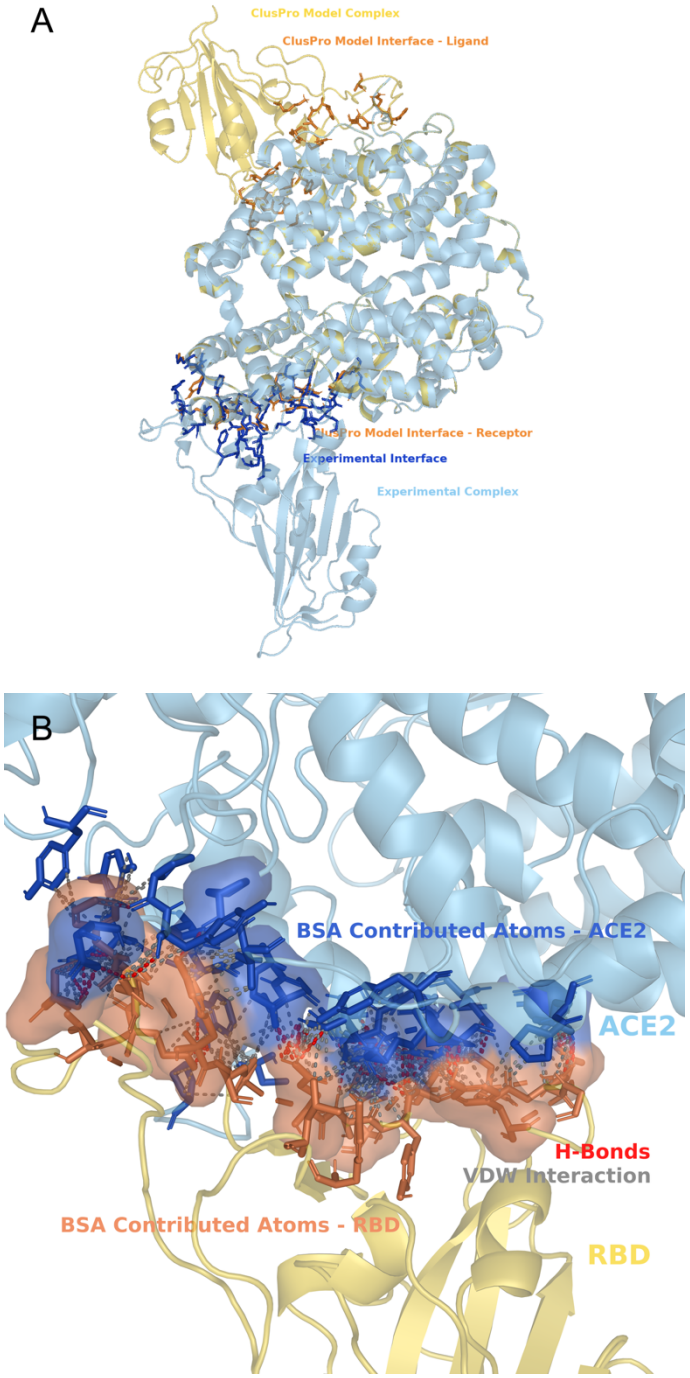


Figure 2. (A) *Superposition of the top-ranked ClusPro model (yellow cartoon) onto the experimental complex (light-blue cartoon), aligned on ACE2* ***(interface Cα RMSD = 37.917 Å)****. The crystallographic ACE2 interface residues are shown as dark-blue sticks, whereas the interface residues of the docked RBD are shown as orange sticks.* ***(B)*** *Close-up view of the ClusPro-predicted interface. ACE2 is depicted as a yellow cartoon, and the RBD is shown as a light-blue cartoon. ACE2 interface residues appear as dark-blue sticks and RBD interface residues as orange sticks.* ***Grey dashed lines*** *indicate atom pairs with* ***van der Waals contacts****,* ***red dashed lines*** *mark* ***hydrogen bonds****, and* ***dark blue or orange surfaces highlight residues contributing*** *to total* ***BSA****.*

1.3

- ***Results***
- Docking results analyses from PDBePISA were listed in *Table 3*.
- ***Trends***
- The experimental ACE2–RBD complex exhibits a smaller buried surface area on ACE2, on RBD, and in total, but a higher number of hydrogen bonds and salt bridges compared to the ClusPro model.
- These values suggest that the docked model forms a deeper, more buried interface, but one that is less chemically optimized. The crystal structure achieves tighter recognition through specific electrostatic and hydrogen-bonding networks, rather than maximizing packing alone.
- ***Possible Reasons***
- First, **rigid-body approximation** in ClusPro prevents backbone and side-chain relaxation, often causing steric over-packing and optimizing BSA, while failing to consider hydrogen-bond geometries well.
- Second, **side-chain flexibility**, which is essential for forming salt bridges and optimized hydrogen-bond networks, is not explicitly sampled during docking and therefore leads to an underestimation of electrostatic interactions.
- Third, **the simplified solvent model** in docking scoring functions cannot fully capture desolvation penalties or water-mediated interactions, both of which contribute substantially to the stability of the experimental ACE2–RBD interface. As a result, the docked interface is geometrically plausible but chemically incomplete.
- ***Reflection on ClusPro docking***
- ClusPro docking combined with PDBePISA analysis provides a valid primary approximation of large protein–protein interfaces, particularly for estimating interface geometry and identifying potential contact regions. However, the method has limited reliability when assessing interaction energetics or detailed chemical specificity. The rigid-body nature of the docking algorithm, combined with the absence of explicit side-chain repacking or solvation effects, leads to a systematic overestimation of buried surface area and a loss of native hydrogen-bond and salt-bridge networks. Consequently, while ClusPro–PISA pipelines can guide hypotheses and screen plausible binding orientations, they cannot replace experimentally determined structures or refined computational protocols involving flexible docking or molecular dynamics relaxation. Hence, docking programs and algorithms remain meaningful yet challenging topics in structural computational biology.

	BSA (ACE2) (Å²)	BSA (RBD) (Å²)	BSA (Total) (Å²)	No. Hydrogen Bonds	No. Salt Bridges
Experimental Structure	*849.1*	*861.0*	*1710.1*	*16*	*9*
Docking Structure	*1033.0*	*1101.9*	*2134.9*	*9*	*0*

Table 3. *PDBePISA interface parameters for the experimental structure (6M0J) and the ClusPro docking model, including buried surface area, number of hydrogen bonds, and number of salt bridges.*

Q2

2.1

- Using 1R4L, we defined pocket residues as ACE2 residues within 4.5 Å of any MLN-4760 (XX5) atom around the catalytic zinc ion (Towler et al., 2004).
- This distance cutoff yields **21 residues** *(Figure 3 A)*, which are listed with their functions in *Table 4*.
- To be more specific (Towler et al., 2004):
  - The catalytic core consists of **His-374**, **His-378,** and **Glu-402**, which coordinate Zn^{2+} and position the inhibitor's zinc-bound carboxylate.
  - 6 (7) nearby residues form a local hydrogen-bond network with MLN-4760. The side chains of **Arg-273**, **His-505,** and **His-345** hydrogen bond to the terminal carboxylate of the inhibitor. The backbone carbonyl of Pro-346 and the N atom of **His-345** lie within hydrogen-bonding distance of the secondary amine group. **Thr-371** contacts the imidazole ring of the 3,5-dichlorobenzylimidazole moiety. The phenolic OH of Tyr515 donates a hydrogen bond to the zinc-bound carboxylate, while the side chain of **Glu-375** approaches the second oxygen of this group and is expected to participate during catalysis.
  - The **remaining pocket residues** define large parts of the S1 and S1′ subsites and provide additional electrostatic and van der Waals contacts.
  - The **catalytic zinc ion** lies in a cleft between the two globular subdomains, where one terminal carboxylate of MLN-4760 displaces the catalytic water and completes the Zn^{2+} coordination sphere. To be more specific, the catalytic zinc ion adopts a five-coordinate geometry, forming three bonds with ACE2 pocket residues (His374, His378, and Glu402) and two additional coordination bonds with the carboxylate group of MLN-4760.
- Together, these residues form a compact zinc-centered binding pocket in which MLN-4760 is anchored by direct metal coordination, a dense hydrogen-bond network, and shape-complementary hydrophobic packing.

Pocket Residue	*Function*
Glu-145	*Electrostatic and van der Waals interaction site*
Asn-149	*Electrostatic and van der Waals interaction site*
Arg-273	*H-bonding interaction site*
Cys-344	*Electrostatic and van der Waals interaction site*
His-345	*H-bonding interaction site*
Pro-346	*H-bonding interaction site*
Thr-347	*Electrostatic and van der Waals interaction site*
Met-360	*Electrostatic and van der Waals interaction site*
Cys-361	*Electrostatic and van der Waals interaction site*
Lys-363	*Electrostatic and van der Waals interaction site*
Asp-368	*Electrostatic and van der Waals interaction site*
Thr-371	*H-bonding interaction site*
His-374	*Zinc coordination sphere*
Glu-375	*H-bonding interaction site (conditional)*
His-378	*Zinc coordination sphere*
Glu-402	*Zinc coordination sphere*
Phe-504	*Electrostatic and van der Waals interaction site*
His-505	*H-bonding interaction site*
Tyr-510	*Electrostatic and van der Waals interaction site*
Arg-514	*Electrostatic and van der Waals interaction site*
Tyr-515	*H-bonding interaction site*

Table 4. *ACE2 residues that form the MLN-4760 binding pocket in 1R4L and their proposed functional roles.*

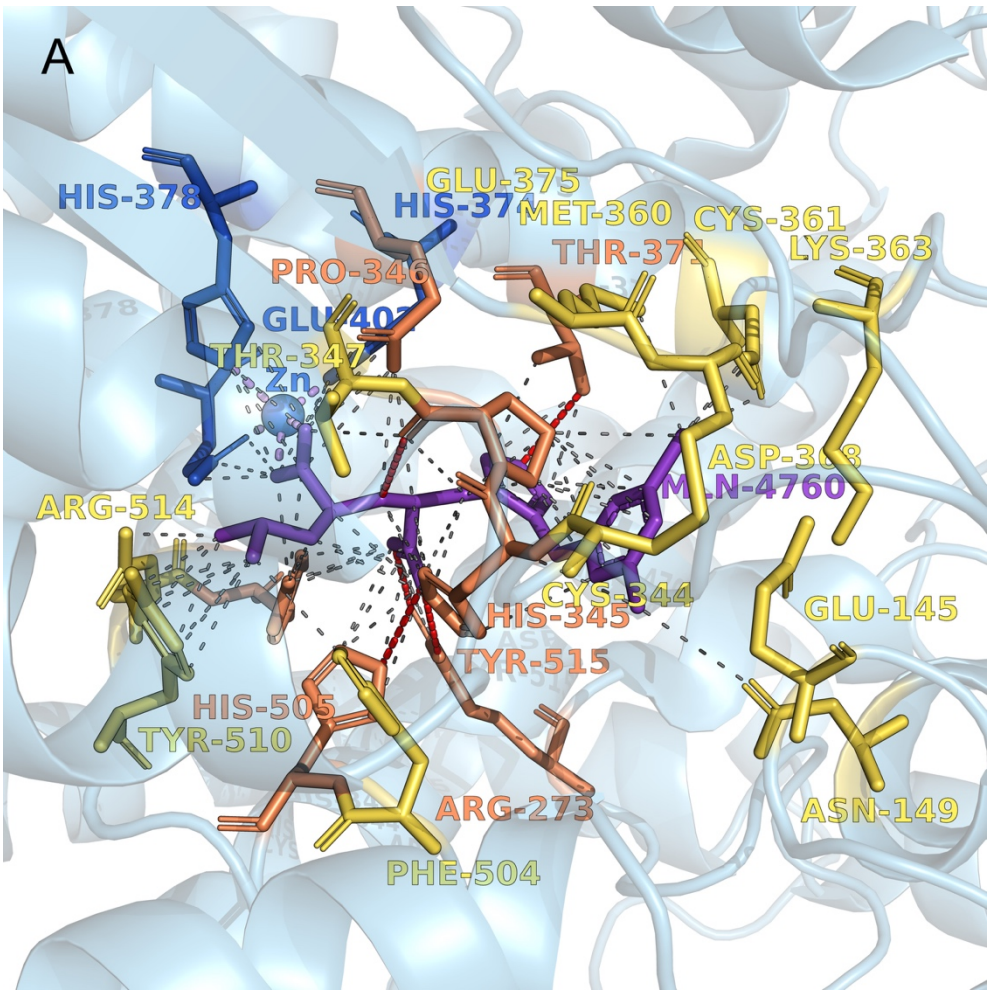


Figure 3. (A) *Close-up view of the MLN-4760 binding pocket in 1R4L.* ***ACE2*** *is shown as a light-blue cartoon. The* ***catalytic Zn^{2+} ion*** *and its coordinating residues* ***(His374, His378, Glu402****) appear in dark blue.* ***MLN-4760*** *is shown as purple sticks at the center of the pocket.* ***Hydrogen-bonding residues (Arg273, His345, Pro346, Thr371, His505, Tyr515)*** *are colored orange, and the* ***remaining pocket residues*** *are colored yellow.* ***Hydrogen bonds*** *are drawn as red dashed lines,* ***electrostatic or van der Waals contacts*** *as grey dashed lines, and a* ***five-coordinate geometry of the zinc ion*** *in pink dashed lines.*

2.2

- ***Docking Preparation***
- Starting from 1R4L, MLN-4760 was removed to generate apo ACE2 models under several solvent and environment settings *(Table 5)*. Zn^{2+} is always retained because MLN-4760 binds by directly coordinating the metal and mimicking the zinc-bound reaction intermediate, making accurate docking impossible without it.
- The ligand was defined by its SMILES string retrieved from PubChem (CID 448281):CC(C)C[C@@H](C(=O)O)N[C@@H](CC1=CN=CN1CC2=CC(=CC(=C2)Cl)Cl)C(=O)O.
- ***Searching Box***
- As metal coordination is central to MLN-4760 binding, the docking search box was centered on the catalytic Zn^{2+} ion at (39, 8, 22) Å. The box size was set to 15 × 15 × 15 Å to sample a relatively large region around the active site while remaining within the practical SwissDock time limit (1 h) *(Figure 4 A)*. All runs used the "Attracting Cavities 2.0" mode with default parameters.
- According to the best orientation RMSD comparison in Table 5, we use the result from the input environment with (*ACE2 + Zn^{2+} + H_2O) for further analysis*
- ***Post Docking Analyses***
- The docking results were summarized as the top five clusters, with their AC scores (docking scores) and SwissParam (ΔG values) reported in *Table 6*.
- Here, poses were **ranked by AC score** from low to high. The AC score combines all energy terms in the SwissDock scoring function and is therefore suitable for ranking poses of a single ligand against a single target. In contrast, the SwissParam score is a fast estimate of binding free energy. It is helpful to compare different ligand–target pairs, but it is not reliable for ranking alternative poses of the same ligand on the same protein.
- Key interactions in the best-scoring pose were then examined in detail *(Figure 5 A)*. A 4.5 Å cutoff around the ligand was used to define the **binding pocket** and to identify **hydrophobic contacts**. A 3.2 Å cutoff was applied to locate potential **hydrogen bonds**, and charged side chains within 10 Å of the ligand were used to screen for **electrostatic interactions**. In addition, ligand atoms within 4 Å of the **Zn^{2+} ion** were searched, but no direct metal–ligand contacts were observed in this pose. Carbon–ring contacts within the hydrophobic set were reclassified as **π- interactions**. The pattern of contacts showed minor differences from the interaction panel displayed by SwissDock, but the overall interaction map was broadly consistent.
- The docked pose was then compared to the experimental inhibitor orientation in 1R4L *(Figure 5 B)*. Superposition of the two MLN-4760 conformations gave an **RMSD of 3.242 Å**, indicating a noticeable shift in binding mode relative to the crystal structure.

Task ID	Environment Components	Best RMSD (Å)	Best AC Score
1	ACE2 + Zn^{2+}	3.354	27.955155
2	ACE2 + Zn^{2+} + H_2O	3.242	27.955155
3	ACE2 + Zn^{2+} + H_2O + Cl^-	3.432	30.294166
4	ACE2 + Zn^{2+} + H_2O + Cl^- + NAGs	3.364	26.215550

Table 5. *SwissDock input systems. All inputs were prepared by removing MLN-4760 and using identical parameters. For each task, the smallest RMSD and corresponding AC Score among all predicted orientations compared to ligand orientation in 1R4L are listed.*

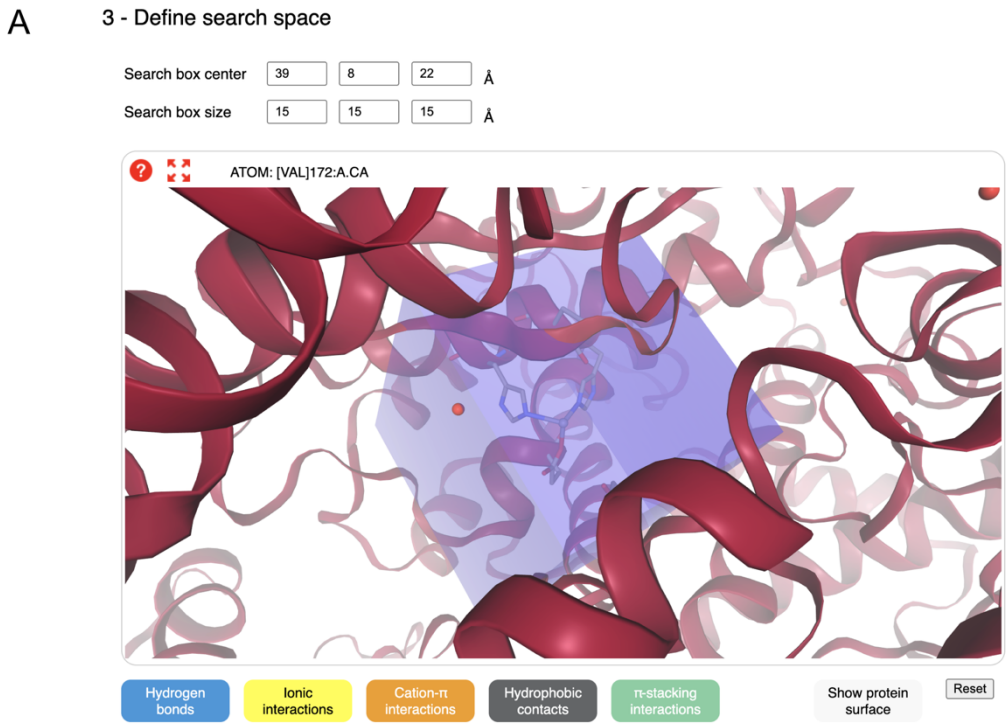


Figure 4. (A) *Search box definition for the docking task with the best agreement to 1R4L (Task ID 2 in Table 5). ACE2 is depicted as a cartoon, with Zn^{2+} represented as a sphere at the center of the box. The cubic search box is outlined, and key contacts in the binding site are indicated to illustrate the region sampled during docking.*

Cluster number	*Cluster member*	*AC Score*	*SwissParam Score (ΔG)*	*Cluster number*	*Cluster member*
0	*1*	*26.036625*	*-7.2493*	*0*	*1*
1	*1*	*26.674111*	*-7.4103*	*1*	*1*
2	*1*	*30.961437*	*-7.3467*	*2*	*1*
3	*1*	*32.060641*	*-7.6245*	*3*	*1*
4	*1*	*33.503090*	*-7.1548*	*4*	*1*

Table 6. *Top five SwissDock clusters for MLN-4760 bound to ACE2, ranked by AC score (from lower to higher). For each cluster, the AC score and the corresponding SwissParam ΔG (kcal/mol) are reported.*

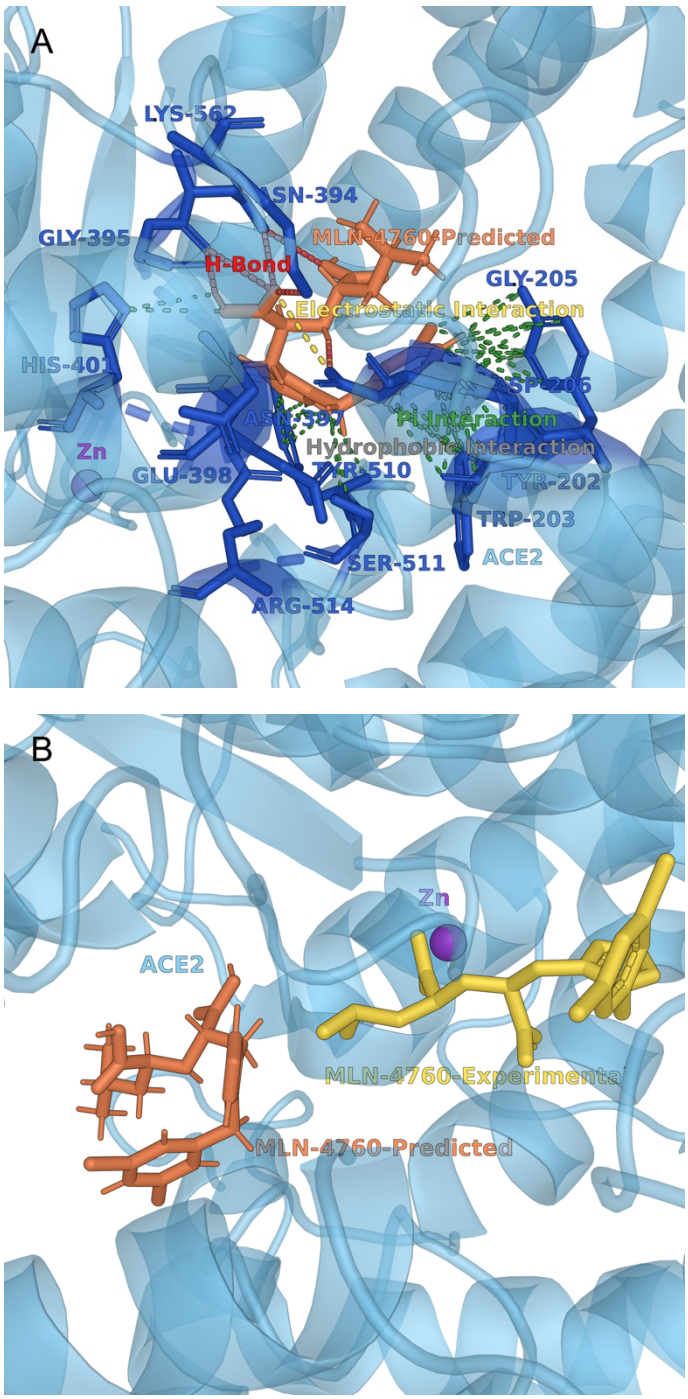


Figure 5. (A) *Close-up view of the predicted MLN-4760 binding pocket and its key interactions. ACE2 is depicted as a light-blue cartoon, with pocket residues represented by blue sticks and the docked MLN-4760 shown in purple sticks.* ***Hydrophobic contacts*** *are indicated by grey dashed lines,* ***hydrogen bonds*** *by red dashed lines,* ***electrostatic interactions*** *by yellow dashed lines, and* ***π- interactions*** *by green dashed lines.* ***(B)*** *Comparison between the experimental and docked orientations of MLN-4760 in the ACE2 active site. ACE2 is shown as a light-blue cartoon, with Zn^{2+} displayed as a purple sphere.* ***The experimental MLN-4760 conformation*** *is depicted as yellow sticks, and the docked pose is shown as orange sticks, illustrating the deviation between the predicted and crystallographic binding modes (RMSD = 3.242 Å).*

2.3

- The experimental ACE2–MLN-4760 complex and the top-ranked SwissDock complex were analyzed with the PLIP webserver to obtain detailed interaction profiles. PLIP identified hydrophobic contacts, hydrogen bonds, π–stacking, π–cation interactions, halogen bonds, salt bridges and metal coordination for both structures *(Table 7; Figure 6 A, B)*.
- Docking MLN-4760 back into ACE2 with SwissDock does not fully reproduce the crystallographic pose of MLN-4760. Several features of the ACE2 active site and the docking protocol can explain this mismatch:
- Incomplete treatment of **Zn^{2+} coordination chemistry**
  - The 1R4L structure displays a well-defined five-coordinate Zn^{2+} center, with His374, His378, and Glu402 binding the metal, and one carboxylate of MLN-4760 occupying the remaining two coordination sites. Standard docking protocols often represent Zn^{2+} as a simple point charge without enforcing the correct coordination number, preferred geometry, or bidentate binding by a carboxylate. As a result, poses that do not place the ligand carboxylate in the metal site can still receive favorable scores, and the predicted orientation drifts away from the crystallographic state.
  - For refinement, consider using a docking protocol that includes explicit metal constraints, such as distance or angle restraints between Zn^{2+} and the ligand's carboxylate oxygens, or a metal-aware scoring term. Alternatively, perform docking with a program that supports metal templates, and then refine the best poses with restrained minimization or **short MD simulation** while keeping Zn–ligand bonds near the crystallographic distances.
- **Ambiguous protonation states** of the active site and ligand
  - The ACE2 pocket contains several protonatable residues (His345, His374, His378, Glu375, Tyr515), and MLN-4760 itself carries two carboxylates and a secondary amine. **Default protonation at "pH 7"** may not match the microstate present in the metal-bound complex. This shifts the local electrostatics and weakens the specific hydrogen-bond pattern that stabilizes the experimental pose.
  - For refinement, consider predicting protonation states with a pK_a tool and preparing several plausible microstates of ACE2 and MLN-4760, including metal-bound charge states. Dock MLN-4760 into each microstate and compare how often the Zn-bound carboxylate and the Tyr515/Arg273/Glu375 hydrogen-bond network are recovered.
- **Limited sampling** of a **flexible ligand**
  - MLN-4760 is a **flexible inhibitor** with multiple rotatable bonds and an extended aromatic system. Within **the restricted time, space, and depth allowed, SwissDock may not fully explore the torsion space required** to align both carboxylates, the central linker, and the dichlorobenzylimidazole ring. The box size and "Attracting Cavities" mode

may further bias sampling towards alternative cavities, so even energetically reasonable poses sit away from the native orientation.
    - For refinement, consider generating a set of low-energy MLN-4760 conformers before docking and docking each conformer separately. Also, try to improve searching time, space, and depth if resource allows.
- **Rigid receptor** and lack of **induced fit**
    - In SwissDock, ACE2 is essentially **rigid**. However, several side chains in the pocket adjust their rotamers to wrap around the inhibitor and support Zn-centered binding. A rigid model cannot capture these subtle side-chain motions, so the scoring function may **favor poses that avoid steric clashes but lose the correct hydrogen bonds and metal geometry**, leading to the observed RMSD between docked and experimental MLN-4760.
    - For refinement, consider refining the best poses with restrained MD or energy minimization in explicit solvent to allow local induced fit around the Zn^{2+} site after primary docking.
- **Solvent** and **scoring** limitations
    - The ACE2 active site contains structural water molecules and experiences a substantial desolvation penalty when binding a charged, metal-binding ligand. **Implicit solvent and generic scoring terms** in SwissDock may not capture water-mediated hydrogen bonds, the cost of desolvating the Zn^{2+} environment, or the balance between burial and electrostatics. In addition, the AC score and SwissParam ΔG are **not calibrated for metal coordination**. They may under-weight correct Zn–carboxylate binding while **over-weighting deeper but chemically less realistic packing**.
    - For refinement, consider retaining conserved water molecules near Zn^{2+} during docking (as we did) and re-ranking the top poses with a more detailed method, such as MM-GBSA or MM-PBSA, on the Zn–ligand complex.
- Overall, protein-ligand docking remains a challenging topic in structural computational biology.

Index	Experimental Structure (1R4L)		Docking Structure	
	Residue	Distance	Residue	Distance
		Hydrophobic Interactions		
1	PRO-346	3.48	TYR-202	3.48
2	PHE-504	3.60	GLU-398	3.93
3	TYR-510	3.44		
4	TYR-510	3.42		
		Hydrogen Bonds		
1	HIS-345	3.22	ASP-206	2.91
2	TYR-515	3.24	ASN-394	3.26
3			ASN-394	2.95
4			ARG-514	3.67
5			ARG-514	3.18
		π-Stacking		
1	HIS-345	5.29	TRP-230	5.05
		π-Cation Interactions		
1	ARG-273	5.61		
		Halogen Bonds		
1	ASP-368	2.74		
		Salt Bridges		
1	ARG-273	3.48	LYS-562	3.69
2	HIS-345	4.52		
3	HIS-374	4.79		
4	HIS-378	4.37		
5	HIS-505	3.81		
		Metal Complexes		
1	HIS-374	2.05		
2	HIS-378	2.31		
3	GLU-402	2.05		
4	XX5 (MLN-4760)	2.04		
5	XX5 (MLN-4760)	2.62		

Table 7. *PLIP-derived interactions between MLN-4760 and ACE2 in the experimental and predicted complexes. For each interaction type, the table lists the interacting ACE2 residues and the corresponding distances (Å).*

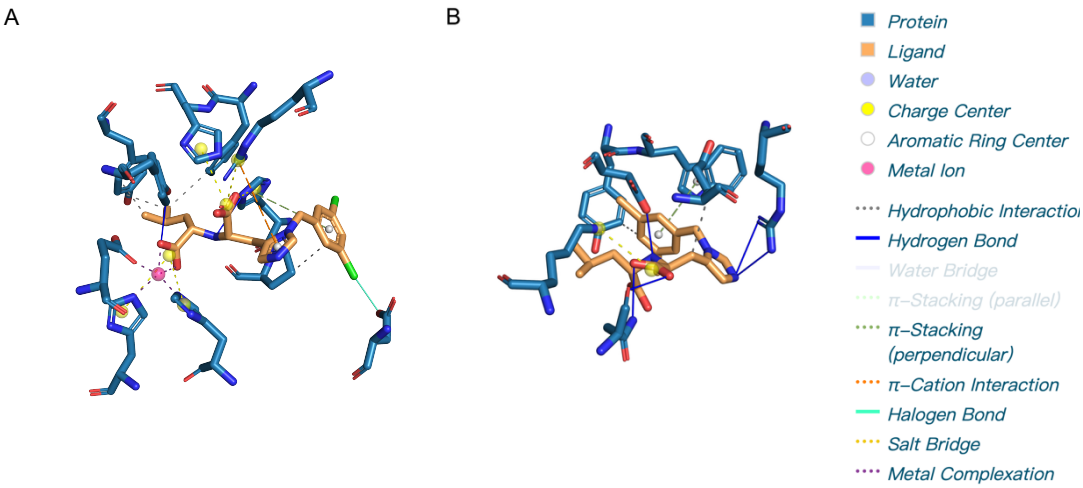


Figure 6. *Overview of interactions between MLN-4760 and the ACE2 binding pocket reported by PLIP.* ***(A)*** *Experimental 1R4L complex.* ***(B)*** *Top-ranked SwissDock complex.*

Reference

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